

Probability : Branch of mathematics that provides the language necessary for situations where uncertainty is involved.

Basic set theory:

Set : Collection of distinct objects known as its "elements".
Finite set : Finite no. of elements.

No. of elements of a set A : $\#A$

Set operations:

$A \cap B$ = the set of elements that belong to both A & B.
(at the same time)

$A \cup B$ = the set of elements that belong to either A or B (or both)

$\emptyset$ = empty set (set with no elements)

$$\# \emptyset = 0$$

① ("omega") $\Omega \neq \emptyset$ Sample space (set of all possible outcomes).
Eg. rolling a die
 $\Omega = \{1, 2, 3, 4, 5, 6\}$
the elements of Ω are called "outcomes".

$\mathcal{F} \subseteq 2^\Omega \rightarrow$ set of events

② Possible events are the subset of Ω
event $\neq$ outcome!

subset of S

element of S

The Probability of an event

$$P: \mathcal{F} \rightarrow [0, 1]$$

Axioms of Probability :

(i) for any $A \subseteq \Omega$, $P(A) \geq 0$

(ii) $P(\Omega) = 1$

(iii) $A_i \cap A_j = \emptyset$ (A_i & A_j are mutually exclusive)

then
 $P(A_i \cup A_j) = P(A_i) + P(A_j)$

Remarks :

(i) The above axioms must always hold.

(ii) When outcomes are "equally likely", then
we define $P(A) = \frac{\#A}{\#S}$ "same chance" of occurring

(iii) All the properties of the probability funcⁿ can be derived from the Axioms.

Propⁿ: (i) $P(A^c) = 1 - P(A)$ for any event

(ii) $P(\emptyset) = 0$

(iii) if $A \subseteq B$, then $P(A) \leq P(B)$.

(iv) for any event A , $0 \leq P(A) \leq 1$.

(v) for generic events A & B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Classical Perspectives of Probability

"equally likely outcomes"

Ex. ① Experiment : Flipping 2 coins

Sample space

$$\Omega = \{\text{ordered pairs of the 2 outcomes}\}$$
$$\{ (H,H), (H,T), (T,H), (T,T) \}$$

$$E_1 = \{\text{both flips come up as head}\}$$

$$= \{ (H,H) \}$$

$$E_2 = \{\text{the 2 flips come up different}\}$$

$$= \{ (H,T), (T,H) \}$$

$$P(E_1) = 1/4 ; P(E_2) = 1/2$$

② Experiment : Repeatedly rolling a die until we first get 6

$$\Omega = \{\text{sequences of numbers bet}^n 1 \& 5 \& \text{ then a 6}\}$$

$$|\Omega| = \infty$$

$$E_1 = \{\text{roll 4 first, get 6 on the 3rd roll}\}$$
$$\{ (4, m, 6) , m = 1, 2, 3, 4, 5 \}$$

Frequentist Interpretation of Probability :

Define probability via a thought experiment.

Ex. Roughly speaking, suppose that we have a die which is weighted, but we don't know how it is weighted.

We could get a rough idea of the probability of each outcome by tossing the die a large no. of times & using the proportion of times that the die gives that outcome to estimate the probability of that outcome.

each 'repetition' is called a "TRIAL".

In the FREQUENTIST INTERPRETATION, $P(E)$: Proportion of times, in repeated & independent trials of an experiment, in which

• E occurs (by "independent trials we mean that they don't influence each other").

Precisely, let $n = \#$ of times the process is performed, (the trial no.)

and
 $m = \#$ of times the outcome A happens out of the first n trials.

Remark ① Notice that m & n stand for different things in this def.ⁿ.

If $\frac{m}{n}$ approaches some value for large n , in the 'frequentist perspective' to probability,

$$P(A) = \lim_{n \rightarrow \infty} \frac{m}{n}$$

Remark 2: In this approach, the probability of an event A, i.e., the no. $P(A) \rightarrow$ can be objectively measured with a sequence of repeated & independent trials of an experiment.

Remark 3: F generalizes C (i.e. if we have a fair dice)
 get same answer

" 4: F works when outcomes are not equally likely, such as the weighted die.

" 5: F is in principle just a 'thought' experiment in some cases.

The experiment can never in practice be carried

out more than once. *
e.g. the probability that Kohli's arg. will go up — "
 " " of going to war with country X. — "

Subjective perspective: Individual's measure of belief that an event will occur.
With this view, it makes sense to talk about events.

Limitations: subjective (individual centric)
" (must obey certain coherence consistency conditions in order to be workable)

Axiomatic Perspective

↓
unifying perspective. Could these coherence conditions be used with any of the above perspectives.
———— " ————

Thursday (18/1)

Independent events: A, B are independent if
 $P(A \cap B) = P(A)P(B)$ (knowing whether one occurs provides no useful information about the other)

Remark: Different from A & B being mutually exclusive / disjoint.
(if A occurs then B cannot occur)

Ex. Experiment: Drawing 1 card at random from a deck of 52 cards.

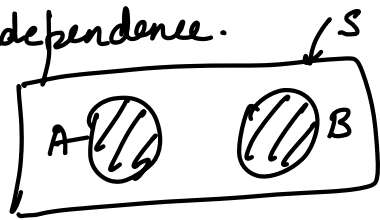
A : Card is an ace
 B : " " a heart } are they independent?

• $P(A) = 4/52 = 1/13$ • $P(B) = 13/52 = 1/4$

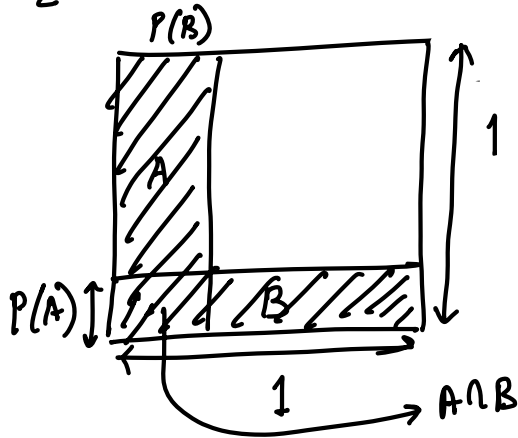
• $P(A \cap B) = P(\text{"the card is the ace of heart"}) = 1/52$

$$\frac{1}{52} = \frac{1}{13} \cdot \frac{1}{4}, \quad A \text{ \& B are independent.}$$

Venn diagram representation of Independence.
Mutually exclusive / disjoint $\leftarrow$



2 events A & B are independent



Interpret area of a set
as the probability of that
set.

$$P(A \cap B) = P(A) \cdot P(B)$$

Note: Independence of the
sets A & B is depicted
through the sets A & B
being orthogonal to each other.

Conditional Probability:

Defⁿ: The conditional probability of A given B is

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (\text{assuming that } P(B) > 0)$$

probability that A happens given that B has happened.

Eg. Tossing a coin 3 times:

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

$$A = \text{head atleast twice} = \{HHH, HHT, HTH, THH\}$$

$$B = \text{the 1st toss lands head} = \{HHH, HHT, HTH, HTT\}$$

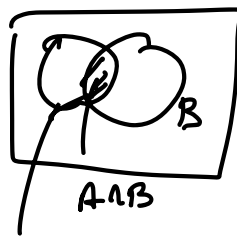
"Win" if A happens

$$P(A) = 4/8 = 1/2$$

$$P(A|B) = 3/4$$

Geometric interpretation of

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



Proportion of B which is occupied by A.

Rem 1: If A & B are independent, then

$$P(A \cap B) = P(A)P(B)$$

$$P(A|B) = P(A)$$

& vice-versa is also true

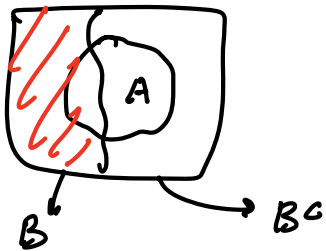
General way to solve a problem. There are different strategies.

(i) try simple & extreme cases (useful in many problems)

(Feynman's strategy)
Write down the problem
think hard
Write the solⁿ

(ii) Useful in statistics → break the problem into simpler problems (more problems but each easier)

Let A & B be generic events in S:



Find $P(A)$?

Break it into pieces using B.

$$A = (A \cap B) \cup (A \cap B^c)$$

mutually exclusive

$$(A \cap B) \cap (A \cap B^c) = \emptyset$$

Therefore, by Axiom 3:

$$P(A) = P(A \cap B) + P(A \cap B^c) \quad (*)$$

But by conditional probability

$$P(A \cap B) = P(A|B) \cdot P(B) \quad \&$$

$$P(A \cap B^c) = P(A|B^c) \cdot P(B^c)$$

hence (*) becomes

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c)$$

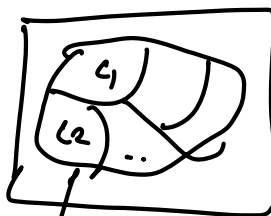
Total probability thm.

Generalized total probability thm.

Defⁿ: The events $C_1, C_2, \dots, C_n$ are called a 'Partition of A'

(i) $C_1 \cup C_2 \cup \dots \cup C_n = A$

(ii) for $i \neq j$, $C_i \cap C_j = \emptyset$



Propⁿ: If $C_1, C_2, \dots, C_n$ is a partition of S.

Then:

$$P(A) = P(C_1) + \dots + P(C_n) \quad (*)$$

If $B_1, B_2, \dots, B_n$ is a partition of S.

let A be a generic event in S

then $A \cap B_1, \dots, A \cap B_n$ is a partition of A.

(since $(A \cap B_1) \cup \dots \cup (A \cap B_n) = A$
for $i \neq j$, $(A \cap B_i) \cap (A \cap B_j) = \emptyset$)

Therefore for $C_i = A \cap B_i$

(*) yields $P(A) = P(A \cap B_1) + \dots + P(A \cap B_n)$

$$\Rightarrow P(A) = P(A|B_1)P(B_1) + \dots + P(A|B_n)P(B_n)$$

Generalized total probability thm.

Ex. We have 2 urns

urn #1: 9 white balls & 1 red ball

urn #2: 1 " " & 4 " "

Experiment: (i) Toss a fair coin

- if heads, choose urn #1

- if tails, choose urn #2

(ii) We then pick a ball at random from the chosen urn.

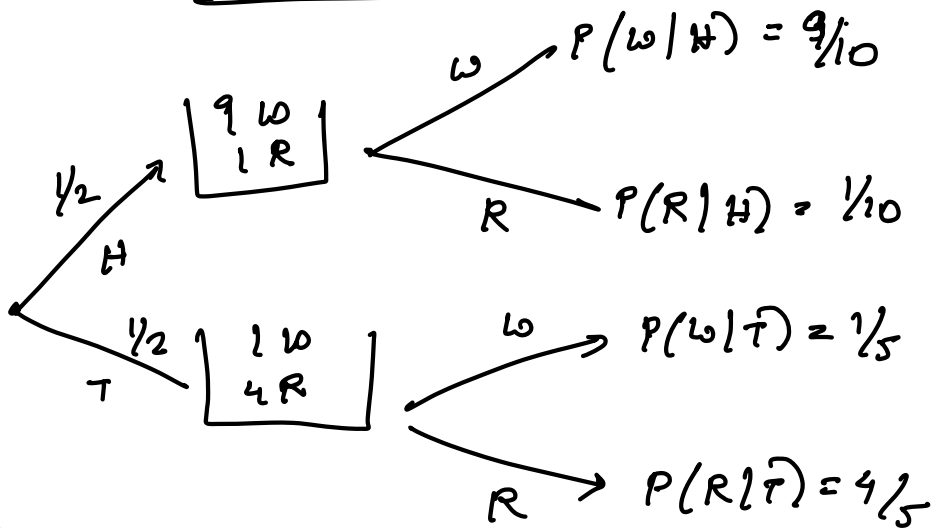
Denote W = "pick a white ball"

H = "coin shows head"

R = "pick a red ball"

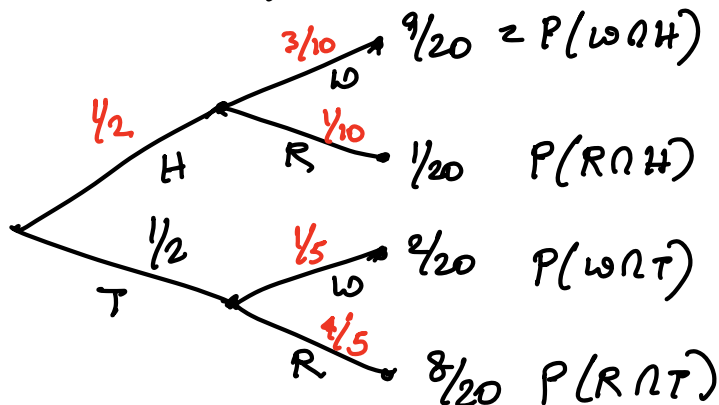
T = "coin shows tail"

Use a TREE DIAGRAM:



So:

$$\begin{aligned} P(W) &= P(W|H)P(H) + P(W|H^c)P(H^c) \\ &= P(W|H)P(H) + P(W|T)P(T) \\ &= \frac{9}{10} \cdot \frac{1}{2} + \frac{1}{5} \cdot \frac{1}{2} = \frac{9}{20} + \frac{1}{20} = \frac{10}{20} \end{aligned}$$



Q. Now assume that at the end, we have picked a white ball.

What are the chances that we tossed heads in the 1st place?

What is $P(H|W)$?

Use Bayes' Rule:

If A & B are events in S , then:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

denominator can be computed using total probability thm.

$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$$

Using Bayes' Rule

$$P(H|W) = \frac{P(W|H)P(H)}{P(W)} = \frac{1/11}{1/11} \quad \text{!} \quad P(H) = 1/2$$

Using generalized total probability thm, one can get generalized Bayes' Rule

If $B_1, B_2, \dots, B_n$ is a partition of S , then

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{P(A)}$$

By Bayes' rule

$$\Rightarrow P(B_i|A) = \frac{P(A|B_i)P(B_i)}{P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n)}$$

(using total probability thm)

This will help in problems with tree diagrams with multiple branches.

E.g. 3 urns

#1 5W, 1R

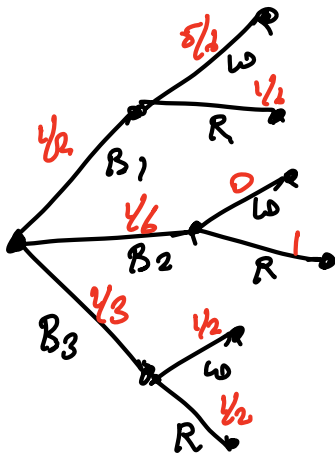
#2 0W, 7R

#3 3W, 3R

Game: Roll a fair die.

• if $B_1 = \{1, 2, 3\}$ choose urn 1
 • if $B_2 = \{4\}$ " " 2
 • if $B_3 = \{5, 6\}$ " " 3

then pick a random marble.



By generalized total probability then

$$P(R) = P(R|B_1)P(B_1) + P(R|B_2)P(B_2) + P(R|B_3)P(B_3)$$

$$= \frac{5}{12}$$

By Bayes rule

$$P(B_2|R) = \frac{2}{5}$$

E.g.-2 Suppose that we are testing for a disease

Patient gets tested for a disease which afflicts 1% of population

Suppose the patient tests positive

Suppose the test (as advertised) is 95% accurate

↳ what does this mean?

if the patient has the disease, then

95% times, the test will correctly report +ve.

same as if the patient does not have the disease, then 95% times, the test will correctly report -ve.

Notation:

D: Patient has disease

T: 4 tests +ve

$$P(T|D) = 0.95 = P(T^c|D^c)$$

What does the patient care? $P(D|T)$

$$P(D|T) = \frac{P(T|D)P(D)}{P(T)} \leftarrow \text{don't know}$$

$$= \frac{P(T|D)P(D)}{P(T|D)P(D) + P(T|D^c)P(D^c)} \approx 0.16$$

0.05

Even though the test is 95% accurate in this sense,
only 16% chance that patients has the disease.